

lar. The next generation of printed sensors will enable applications ranging from human-machine interfaces to environmental sensing. The IDTechEx report predicts the market for fully printed sensors to reach \$7.6 billion by 2027.

Printed sensors may have a very simple structure with only a few electrodes, while others are much more complex requiring deposition of multiple layers. What they have in common is the capability to be manufactured on plastic substrates, which offer advantages in terms of mechanical flexibility, thinness and weight reduction.

MEMS. Microelectromechanical systems (MEMS) are devices characterised both by their small size and the manner in which these are made. These are made up of component sizes between 1 and 100 micrometres. The most notable elements are microsensors and microactuators.

MEMS devices can vary from simple structures to extremely complex electromechanical systems with multiple moving elements under the control of integrated microelectronics. In other words, MEMS sensor is a precision device in which mechanical part and micro sensors along with a signal-conditioning circuit are fabricated on a small piece of silicon chip.

Generally, MEMS consist of mechanical microstructures, microactuators, microsensors and microelectronics in one package. Fig. 7 shows the block diagram of a MEMS device.

Microsensors detect changes in a system's environment by measuring thermal, chemical, electrical or mechanical information. These variables are processed by microelectronics and then microactuators act according to the changes in the environment.

Some common types of MEMS sensors available in the market are:

1. **MEMS accelerometers.** These are used to measure static or dynamic force of acceleration. The major categories are silicon capacitive, piezoresistive and thermal accelerometers.

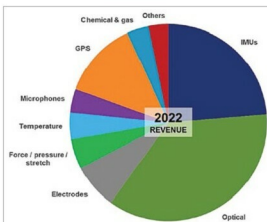


Fig. 5: Wearable chart (Courtesy: www.idtechex.com)

Renesas 8.48 MP High-Sensitivity CMOS Image Sensor for 4K Video Network Cameras



Fig. 6: Image sensor (Courtesy: www.renesas.com)

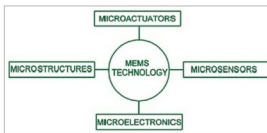


Fig. 7: Block diagram of a MEMS device

MEMS accelerometers are used in smartphones for various controls including switching between landscape and portrait modes, anti-blur capture and pocket-mode operation.

2. **MEMS gyroscopes.** These detect the angular rate of an object. MEMS gyros are used for vehicle stability control with a steering-wheel sensor and rollover detection.

3. **MEMS pressure sensors.** These sensors measure three types of pressures: gauge, absolute and differential pressure. The sensor is integrated with a diaphragm and a set of resistors on integrated chips so that pressure is detected as a change in resistance.

These sensors are used in automotive, industrial, medical, defence and aerospace applications. In automotive systems these are widely used in oil pressure sensor, crash detection, fuel-tank vapour pressure monitoring, exhaust gas recirculation, engine management system, etc.

4. **MEMS magnetic field sensors.** These sensors detect and measure magnetic fields, and find use in position sensing, current detection, speed detection, vehicle detection, space exploration, etc.

5. **Fluxgate sensors.** Fluxgate sensors are used to measure DC or low-frequency AC magnetic field. These find many applications like space research, geophysics, mineral prospecting, automation and industrial process control. MEMS-based fluxgate sensors score over other fluxgate sensors owing to their less power consumption, small size and better performance.

NEMS. Nanoelectromechanical systems (NEMS) are a class of devices like MEMS but on the nanoscale. These are the next miniaturisation step after MEMS devices. Nanoresonators and nanoaccelerometers are examples of NEMS. Usually, NEMS rely on carbon-based materials, including diamond, carbon nanotubes and graphene. One of their most promising applications is the combination of biology and nanotechnology. Nanoresonators would find application in wireless communication technologies, while nanomotors might be used in nanofluidic pumps for biochips or sensors.

Sensor materials

The exact sensor materials depend on their type and application. For example, digital, analogue, proximity and image sensors have their own materials, structures, fabrication techniques and packaging. Here we cover materials and fabrication techniques mainly for the latest sensors like MEMS commonly adopted by most manufacturers. Materials used in electronics can play an active or passive role. Some materials play both roles.

Passive materials. These are used